Tank Test Procedure

27FEB26

Document Information

Document Title	Test Tank Procedure
Document ID	TP-001
Version	1.1
Author(s)	OP
Approver(s)	*Whomever*
Date	28FEB26

Revision

Version	**Date**	**Author**	**Description of Changes**
1.0	27FEB26	OP	Initial Draft
1.1	28FEB26	OP	Adjustments to Format/Test Procedure
1.2	01MAR26	GR	Modified the procedure to include mass measurements
1.3	03MAR26	MS, HK	Adding damage modes and sweeps

Contents

1 Motivation

Tank testing will confirm the digital twin's data outputs, as well as determine the percent effect of variable wing and rudder damage on those data outputs. This will both finalize critical system and functional requirement requirement data needs, as well as begin the characterization of wing and rudder damage in the fault database.

2 Scope

The intention is for this test to occur on Thursday at 1230 at the UW School of Oceanography Test Tank on 5 March 2026. It is intended to integrate with SSC operations and minimize the impact on SSC scheduling. The test should conclude by 1700 on 5 March 2026.

3 Test Environment/Safety/

i Test Environment

The UW School of Oceanography Test Tank is a 7.3 m x 3 m x 3.6 m, 23,283 gallon, room temperature tank covered by 9 lids allowing forklift access. The lids are lifted out with a 5 ton hoist to provide access to the tank. The same hoist is used for placing equipment in the test tank.

ii Safety

Due to the test tank lids and hoist, there is a present crush hazard present during the test. The test tank itself presents a drowning hazard. The Seaglider presents an electrocution and burn hazard if damaged. AA Ocean Capstone students should follow all UW School of Oceanography and SSC staff instructions to prevent risk of injury or death.

4 Roles and Responsibilities

1. ______ **Videographer:** Collects video data for later analysis using camera of choice
2. ______ **Test Lead:** Coordinates with SSC and organizes Capstone members
3. ______ **Data Lead:** Collects Seaglider data from SSC team for analysis
4. ______ **Test Members:** Any Capstone members assisting with or observing test

5 Test Procedure

i Equipment

- ☐ Laptop
- ☐ Camera
- ☐ Tripod
- ☐ Data Recording Device (laptop/USB Drive)
- ☐ Note Taking Device
- ☐ Scale
- ☐ Closed Toe Shoes (for crush hazards)
- ☐ Hardhats *as required*
- ☐ Life vests *as required*
- ☐ Hacksaw *as required*

ii Seaglider Control Limits

Sweep	Max	Min	Center
Pitch	3.3 cm (aft)	-8.3 cm (fwd)	2850 (AD)
Roll	$50°$ (stbd)	$-50°$ (port)	1991 (AD)
VBD	548.2 (cc)	-300.5 (cc)	2735

Table 1: Min, Max, and Center position

iii Test Scenarios and Damage Modes

1 Nominal (No Damage Modes)

1. 100% of wing attached
2. 100% of rudder attached

2 Off Nominal (Damage Modes)

1. 66.6% of wing attached
2. 33.3% of wing attached
3. 0% of wing attached
4. 66.6% of rudder attached
5. 33.3% of rudder attached
6. 0% of rudder attached

3 Pitch Position Sweep

All nominal and off nominal cases will perform this pitch sweep.

1. Move battery pack from the neutral center point to the halfway point of the maximum forward position.
2. Once Seaglider has stopped moving, ensure data collection.
3. Move the battery pack back to the neutral center point.
4. Repeat step 2.
5. Move the battery pack to the maximum forward position.
6. Repeat step 2.
7. Move the battery pack back to the neutral center point.
8. Repeat step 2.
9. Move the battery pack from the neutral center point to the halfway point of the minimum position.
10. Repeat step 2
11. Move the battery pack back to the neutral center point.
12. Repeat step 2
13. Move the battery pack to the minimum position.
14. Repeat step 2.
15. Move the battery pack to the neutral center point

4 Roll Position Sweep

All nominal and off nominal cases will perform this roll sweep.

1. Move battery pack from the neutral center point to the halfway point of the maximum control position.
2. Once Seaglider has stopped moving, ensure data collection.
3. Move the battery pack back to the neutral center point.
4. Repeat step 2.
5. Move the battery pack to the maximum control position.
6. Repeat step 2.
7. Move the battery pack back to the neutral center point.
8. Repeat step 2.
9. Move the battery pack from the neutral center point to the halfway point of the minimum control position.
10. Repeat step 2

11. Move the battery pack back to the neutral center point.
12. Repeat step 2
13. Move the battery pack to the minimum control position.
14. Repeat step 2.
15. Move the battery pack to the neutral center point

5 VBD Position/Volume Sweep

All nominal and off nominal cases will perform this VBD sweep.

1. Control VBD to the neutral buoyancy control position.
2. Deflate VBD from the neutral buoyancy point to the halfway point of maximum control position
3. Once Seaglider has stopped moving, ensure data collection.
4. Inflate VBD to the neutral buoyancy control position.
5. Repeat step 3
6. Deflate VBD to the maximum control position
7. Repeat step 3
8. Inflate VBD to the neutral buoyancy control position.
9. Repeat step 3
10. Inflate the VBD from the neutral buoyancy point to the halfway point of the minimum control position
11. Repeat step 3
12. Deflate the VBD to the neutral buoyancy control position.
13. Repeat step 3
14. Inflate the VBD to the minimum control position.
15. Repeat step 3
16. Deflate the VBD to the neutral buoyancy control position.

6 Equation form of the Test sweeps

Input parameters from Table 1 for maximum, minimum, and center constants into the following matrices in order to iteratively calculated the required control position commands to perform at said iteration.

$$C = \begin{bmatrix} c_{pitch} \\ c_{roll} \\ c_{VBD} \end{bmatrix} \begin{pmatrix} cm \\ deg \\ cc \end{pmatrix}$$

With these constants, they can be compiled into the control position change matrices, which will change based on whether the test's desired control position is a maximum or minimum test. Pitch tests will use Δx_{pitch}, Roll tests will use Δx_{roll}, and Buoyancy tests will use Δx_{VBD} as their Δx in the iterative function for the given test.

$$\Delta x_{pitch} = \begin{bmatrix} \frac{1}{2}(Pitch_{max/min} - c_{pitch}) \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta x_{roll} = \begin{bmatrix} 0 \\ \frac{1}{2}(Roll_{max/min} - c_{roll}) \\ 0 \end{bmatrix}$$

$$\Delta x_{VBD} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2}(VBD_{max/min} - c_{VBD}) \end{bmatrix}$$

$$\sum_{i=0}^{n=2} C + \Delta x \cdot i$$

Where the first iteration, $(i = 0)$, is the neutral state at the beginning of the maneuver, the second iteration, $(i = 1)$,the halfway point from center to the maximum or minimum position depending on the test being performed, and the third iteration, $(i = 2)$, is the final position at either the minimum or maximum control position depending on the test being performed. The final output of each iteration is the entire desired control state of the Seaglider control parameters.

iv Procedure

1. _______ Arrive at Test Tank Facility (1501 NE Boat St, Seattle, WA) at least 15 minutes prior to test
2. _______ Meet with SSC or Test Tank staff to enter
3. _______ Wear suitable PPE and follow any safety rules required by the SSC and the Test Tank staff
4. _______ Assist SSC or Test Tank staff with setup of Seaglider or Test Tank
5. _______ Setup videography equipment to capture full range of motions expected by Seaglider
6. _______ Setup data transfer or recording device to record data from Seaglider
7. _______ Using the scale, weigh the Seaglider body without any control surfaces attached and record the mass
8. _______ Using the scale weigh the Seaglider control surfaces and record their masses.
9. _______ Install rudders and wings according to the standard configuration of the Seaglider

Mass of Seaglider (kg)	

Control Surface	Mass (kg)
Wing 1	
Wing 2	
Rudders	

Total Mass of Seaglider (kg)	

10. _______ Using the scale, weigh the Seaglider with all control surfaces attached and record the total mass
11. _______ Confirm with SSC or Test Tank staff that test may proceed
12. _______ Perform nominal test scenario with buoyancy, roll, and pitch
13. _______ Confer with SSC staff maximum range on above maneuvers: _______cc buoyancy, _______degrees roll, _______degrees pitch
14. _______ Ensure data and video quality is acceptable before continuing
15. _______ Remove Seaglider from Test Tank and retrofit with simulated damage
16. _______ Using the scale, weigh the damaged control surface and record its mass
17. _______ Remove the attached control surface and install the damaged control surface in place of that part
18. _______ Using the scale, weigh the Seaglider in the damaged configuration and record the total mass
19. _______ Repeat steps 11-18 for all damage modes
20. _______ Ensure data and video is stowed securely
21. _______ Assist SSC or Test Tank Staff with clean-up as requested
22. _______ Ensure all Capstone equipment is stowed and secured
23. _______ Exit Test Tank premises and upload data

v Expected Results

It is expected that the Seaglider will perform rolls, pitches, and buoyancy maneuvers differently with damaged surfaces than without. The damage cases are expected to cause higher rates of oscillatory roll and pitch maneuvers in the given test sections. Other expected observations in damage cases include abnormal data changes from Seaglier orientation disturbances in otherwise unaffected data sets. Examples of this include only buoyancy data changing in a nominal buoyancy maneuver, but both buoyancy and roll data change in an off-nominal buoyancy maneuver. What data is affected and the exact percentage difference is to be determined for each fault individually from it's collected data file.

vi Actual Results

Affected Item	**% Control Surface Span Remaining**	**Mass (kg)**	**Expected Results**	**Notes**	**P/F**
Port Side Wing / Top Rudder	100 Nominal		Data and video of acceptable quality and test cleared for simulating damage.		☐ P☐ F
Port Side Wing / Top Rudder	66 Off Nominal		Maneuvers cause detectable orientation shift.		☐ P☐ F
Port Side Wing / Top Rudder	33 Off Nominal		More noticeable orientation shift and faster roll and pitch rates.		☐ P☐ F
Port Side Wing / Top Rudder	0 Off Nominal		Greatest orientation shift and fastest roll and pitch rates.		☐ P☐ F

6 Acceptance Criteria

1. (Yes/No) Data collected showed noticeable difference in nominal/off-nominal conditions
2. (Yes/No) Video post-analysis confirms data collected
3. (Yes/No) Test was concluded safely
4. (Yes/No) SSC pilots are capable of reproducing maneuvers from test in real-world scenarios

7 Test Acceptance

By signing below, the UW AA Ocean Capstone Test Lead and Approver confirm that the test procedure was executed in the designated enviornent

______________________________ ______________________________

Test Lead Signature & Date **Approver Signature & Date**

8 Appendix

i Test Data

ii Glossary

iii Reference